Functional Annotation Report: Human RTL4 (Retrotransposon Gag-like Protein 4)

Gene symbol: RTL4 (synonyms: SIRH11, ZCCHC16, MAR3, MART4) **UniProt:** Q6ZR62 · **Ensembl:** ENSG00000187823 · **Chromosome:** X **Organism:** *Homo sapiens* (Human) · **Protein length:** 310 aa

Summary

RTL4 is not an enzyme, transporter, or classical signaling receptor. It is a "domesticated" retroviral capsid (Gag) gene — one of 11 eutherian Retrotransposon Gag-like (RTL)/sushi-ichi retrotransposon homolog (SIRH) genes captured from an ancient LTR retrotransposon/metavirus. The human protein retains the structural hallmarks of a retroviral Gag: a PEG10-RTL1 C-capsid domain (IPR063300/PF30901) and an intact nucleocapsid-like CCHC zinc "knuckle" (residues ~280–293) embedded in a basic, RNA-binding C-terminus. Rather than functioning intracellularly as a capsid-assembly scaffold for a virus, RTL4 has been repurposed by eutherian mammals into a **secreted, microglia-derived factor that supports the brain noradrenergic (locus coeruleus) system**.

The most precise functional evidence comes from mouse genetics. Targeted knockout of the mouse ortholog *Sirh11/Zcchc16* produces specific cognitive deficits — impaired attention, increased impulsivity, and impaired working/short-term spatial memory — and prefrontal-cortex microdialysis shows a selective deficit in **noradrenaline (NA) recovery** (but not dopamine) after depolarizing stimulation. A Venus knock-in reporter demonstrates that the RTL4 protein is **secreted**, appears postnatally in noradrenaline-innervated brain regions (midbrain, hypothalamus, hippocampus, amygdala), is produced by **microglia**, and is dynamically regulated by noradrenergic/arousal state (upregulated by the β-adrenergic agonist isoproterenol, sustained by the NA-reuptake inhibitor milnacipran, reduced by anesthesia). Together these data define RTL4 as a secreted modulator that promotes NA reuptake/noradrenergic network function during neonatal critical periods of environmental adaptation.

Human relevance is established genetically: RTL4/ZCCHC16 is **X-linked**, and rare hemizygous loss-of-function in males has been implicated in **autism spectrum disorder (ASD)**, consistent with the broader finding that complete X-linked knockouts contribute meaningfully to male ASD risk. Because RTL4 lacks a classical signal peptide yet is secreted, its route to the extracellular space is most plausibly an **unconventional, capsid/extracellular-vesicle (EV)-mediated pathway**, inferred from its better-studied Gag-derived paralogs Arc and PEG10, which self-assemble into virus-like capsids, encapsidate RNA, and are released in EVs via the multivesicular-body pathway. A major caveat is that human RTL4 is expressed at very low, near-threshold levels, so most mechanistic detail derives from mouse models and from structural/evolutionary inference.

Key Findings

Finding 1 — RTL4 is a eutherian-specific, X-linked domesticated LTR-retrotransposon Gag gene

RTL4 (UniProt Q6ZR62; 310 aa in human) is one of **11 sushi-ichi retrotransposon homolog (SIRH)/Retrotransposon Gag-like (RTL) genes** that eutherian mammals derived from a captured retrovirus/metavirus. It is X-linked and encodes a **CCHC-type zinc-finger protein with high sequence identity to the LTR-retrotransposon Gag protein** (PMID: 26402067). UniProt annotates a single CCHC-type zinc finger (residues 278–295); InterPro assigns the PEG10-RTL1 C-capsid domain (IPR063300/PF30901) together with the CCHC zinc finger (IPR001878).

A distinctive evolutionary signature underlines its functional importance: RTL4 is **"highly conserved in three out of the four major groups of the eutherians — euarchontoglires, laurasiatheria and afrotheria — but is heavily mutated in xenarthran species such as the sloth and armadillo"** (PMID: 26402067). This lineage-specific pattern — maintained where it is useful, degraded where it is not — is a classic hallmark of a domesticated gene under selection. The family context is firm: **"Eutherians have 11 retrotransposon Gag-like (RTL)/sushi-ichi retrotransposon homolog (SIRH) genes presumably derived from a certain retrovirus"** (PMID: 37892118).

Interpretation: RTL4's identity is unambiguous and matches the UniProt record. It is a domesticated Gag — a structural/RNA-binding protein scaffold — not a metabolic enzyme.

Finding 2 — RTL4/SIRH11 is required for normal cognition via the brain noradrenergic system

The sharpest mechanistic evidence is loss-of-function genetics in mouse. **"Gene targeting of mouse Sushi-ichi-related retrotransposon homologue 11/Zinc finger CCHC domain-containing 16 (Sirh11/Zcchc16) causes abnormal behaviors related to cognition, including attention, impulsivity and working memory"** (PMID: 26402067).

The neurochemical basis is specific to noradrenaline. Using prefrontal-cortex microdialysis, **"the recovery rate of noradrenaline (NA) was reduced compared with dopamine (DA) after perfusion of high potassium-containing artificial cerebrospinal fluid in knockout (KO) mice"** (PMID: 26402067). The selectivity for NA over DA (two catecholamines sharing much of their synthesis pathway) points to a role specifically in the noradrenergic system — consistent with the locus coeruleus, the brain's principal source of NA. Critically, **"Sirh11/Zcchc16 is the first SIRH gene to be involved in brain function, instead of just the placenta"** (PMID: 26402067), distinguishing RTL4 from placenta-acting SIRH genes such as PEG10/SIRH1 and PEG11/RTL1.

Interpretation: RTL4 supports noradrenergic signaling needed for attention, impulse control, and working memory. The behavioral triad (attention/impulsivity/working memory) is itself the classic profile of NA-dependent executive function.

Finding 3 — RTL4 is a secreted microglial protein that responds to noradrenaline

A Venus knock-in (Rtl4-Venus) reporter resolves the protein's localization and regulation. The RTL4–Venus fusion is a **secreted protein**, detected postnatally in **midbrain, hypothalamus, hippocampus, and amygdala** — all noradrenaline-innervated regions — with peak intensity during **neonatal critical periods of adaptation to novel environments**. Secretion tracks noradrenergic/arousal state: it is **upregulated by the β-adrenergic agonist isoproterenol**, decreased by anesthesia, and maintained by the **NA-reuptake inhibitor milnacipran**, implicating RTL4 in NA reuptake. Mixed glial cultures identify RTL4 as a **microglial gene**, with secretion responding rapidly to isoproterenol.

The secondary literature summarizes this as: **"SIRH11/RTL4 responds to noradrenaline and is thought to contribute to stress-responsive brain functions"** and notes that **"Several members of this group function in microglia"** (PMID: 42511665).

Interpretation: RTL4 operates **extracellularly**, released by microglia in response to noradrenergic/arousal cues, and feeds back on the noradrenergic system (favoring NA reuptake/network maintenance) during early-life adaptation to novelty. This places its site of action in the extracellular space of noradrenergic circuits rather than inside the secreting cell.

Finding 4 — Human loss-of-function of X-linked RTL4/ZCCHC16 is implicated in autism spectrum disorder

Human genetic evidence connects RTL4 to disease. Exome sequencing of ASD cohorts established that rare complete (hemizygous) X-linked loss-of-function knockouts are significantly enriched in affected males: **"Outside the pseudoautosomal regions on the X chromosome, we similarly observe a significant 1.5-fold increase in rare hemizygous knockouts in males, contributing to another 2% of ASDs in males"** (PMID: 23352160). RTL4/ZCCHC16 is an X-linked gene within this framework, and a family carrying a rare nonsense mutation in RTL4/ZCCHC16 produced ASD in a male proband (Ishino et al., 2024); additional ZCCHC16 frameshift/mutation cases with intellectual disability and autism have been reported.

The mouse KO recapitulates ASD-relevant endophenotypes — increased impulsivity, impaired short-term spatial memory, failure to adapt to novel environments, and delayed frontal-cortex NA recovery — providing cross-species convergence between the human genetics and the mechanistic model.

Interpretation: RTL4 is a credible X-linked ASD risk gene whose disease relevance is mechanistically anchored to the noradrenergic deficits seen on gene loss.

Finding 5 — RTL4 is a domesticated Metaviridae capsid gene under purifying selection with signatures of rapid evolution

Phylogenomic analysis of a complete human genome plus 25 vertebrate genomes shows the SIRH/RTL genes are **domesticated capsid (CA) genes from Metaviridae** (retrovirus-like LTR retrotransposons), arising from at least five independent domestication events. The family shows a mix of selective regimes: domestication gave **"three universally retained single-copy genes evolving under purifying selection and two gene families unique to placental mammals, with multiple members showing evidence of rapid evolution"** (PMID: 38507667). Within the family, **"In the SIRH/RTL family, we find diverse amino-terminal domains"** (PMID: 38507667).

This is fully consistent with RTL4's molecular architecture: a retained PEG10-RTL1 C-capsid domain plus a Gag-nucleocapsid CCHC zinc finger, and its documented eutherian-lineage-specific structural variation (conserved in 3 of 4 superorders, degraded in Xenarthra).

Interpretation: RTL4's capsid core is conserved (purifying selection on the structural module) while its N-terminal/lineage-specific regions diversify rapidly — a pattern that may underlie species-specific tuning of its brain function.

Finding 6 — RTL4 retains a canonical nucleocapsid zinc-knuckle but no signal peptide, implying capsid/EV-mediated unconventional secretion

Direct sequence analysis of human RTL4 (Q6ZR62) reveals a **canonical retroviral Gag nucleocapsid zinc knuckle** with exact C-X2-C-X4-H-X4-C spacing (CLYCSQSGHFTRDC, residues 280–293) embedded in a **basic C-terminal region** (7 Lys/Arg within the last 50 residues) — the hallmark single-stranded-RNA-binding module of retroviral Gag. There is **no classical N-terminal signal peptide** (UniProt annotates none; the N-terminal 60 residues lack a hydrophobic h-region, max Kyte-Doolittle 19-aa window = 0.86). Yet the endogenous RTL4–Venus protein is experimentally secreted.

The resolution of this apparent paradox comes from RTL4's paralogs. The neuronal Gag-derived protein **Arc "self-assembles into virus-like capsids that encapsulate RNA"** and **"Endogenous Arc protein is released from neurons in extracellular vesicles that mediate the transfer of Arc mRNA into new target cells"** (PMID: 29328916). Mechanistic work shows Arc capsids are secreted through the endosomal–multivesicular-body (MVB) pathway, with PI3P mediating capsid assembly/secretion (PMID: 39178227).

Interpretation: RTL4's retained capsid + CCHC architecture confers a latent capacity for capsid assembly and RNA encapsidation, and its signal-peptide-independent secretion is most parsimoniously explained by an **Arc/PEG10-like EV/MVB route**. Whether RTL4 actually packages and transfers RNA between cells is an open, testable question.

Finding 7 — Human RTL4 is expressed at very low levels, with cell-type-enhanced expression including noradrenaline-producing adrenal medulla

Human Protein Atlas data classify RTL4 RNA tissue specificity as **"Not detected"** in bulk tissues and in single-nucleus brain data, confirming very low/near-threshold expression (consistent with reports of "very low expression in the brain"). At single-cell resolution it is **"Cell type**

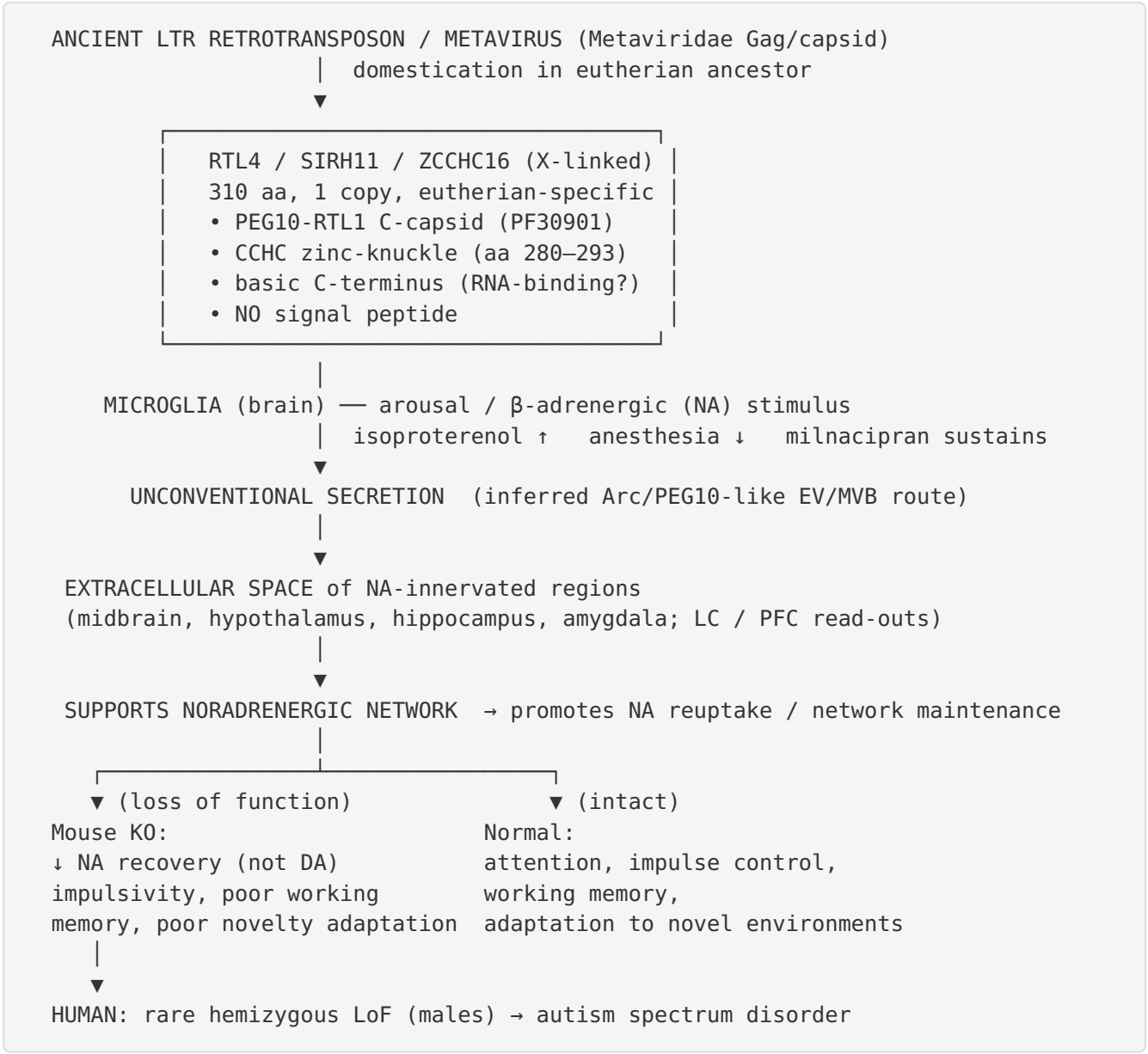
enhanced," detected in a restricted set: pituitary stem cells (nCPM 50.2), pancreatic duct cells (30.1), hepatocytes (28.4), cholangiocytes (15.7), **adrenal medulla cells (10.6)** and Sertoli cells (7.0). Antibody-based protein detection is "Not detected."

The presence of RTL4 in **adrenal medulla cells** — the body's major catecholamine (adrenaline/noradrenaline)-producing endocrine cells — independently reinforces the noradrenergic link seen in brain. Notably, the HPA antibody-based immunofluorescence (cell line) lists nucleoplasm/plasma membrane/cytosol, which **conflicts** with the gold-standard mouse Rtl4-Venus knock-in showing a secreted protein; the discrepancy likely reflects antibody specificity/overexpression artifacts at very low endogenous abundance.

Interpretation: Human RTL4 is a very-low-abundance, cell-type-restricted gene. Its detection in catecholaminergic adrenal medulla cross-validates the catecholamine/noradrenaline theme while also explaining why direct human mechanistic data are scarce.

Mechanistic Model / Interpretation

RTL4 can be understood as a **retroviral capsid protein repurposed into a secreted neuromodulatory factor of the noradrenergic system.** The following schematic integrates the findings:



Functional class. RTL4 is best classified as a **structural/RNA-binding Gag-derived protein acting as a secreted signaling factor** — not an enzyme (no catalytic machinery), not a transporter, and not a classical receptor/ligand with a defined binding partner. Its "substrate" in the retroviral sense would be single-stranded RNA (via the CCHC knuckle), but whether RNA binding/transfer is part of its physiological brain function is not yet demonstrated.

Primary pathway. The clearest pathway connection is to the **noradrenergic/locus-coeruleus arousal system**. RTL4 sits in a feedback relationship with noradrenaline: NA/β-adrenergic signaling drives its secretion from microglia, and RTL4 in turn supports NA reuptake/ noradrenergic network function. This situates RTL4 within neuroimmune–neuromodulatory crosstalk during postnatal critical periods.

Site of action. Extracellular — in the interstitial space of noradrenaline-innervated brain regions. The secreting cell is the microglial cell; the functional target is the noradrenergic circuitry.

Property	RTL4 annotation	Primary evidence
Molecular identity	Domesticated retroviral Gag/capsid protein (CCHC Zn-knuckle + C-capsid)	Sequence/domain analysis; 26402067, 38507667
Enzymatic activity	None known	No catalytic domains
Ligand/substrate	ssRNA (inferred, via CCHC knuckle); physiological target undefined	Structural inference; Arc paralog 29328916
Cellular source	Microglia (brain); adrenal medulla & other epithelia (periphery)	42511665; Human Protein Atlas
Localization of action	Secreted → extracellular space of NA-innervated regions	Rtl4-Venus knock-in (Ishino 2024)
Pathway	Noradrenergic / locus coeruleus arousal system	Microdialysis NA deficit 26402067
Secretion route	Unconventional, signal-peptide-independent (inferred EV/MVB)	No signal peptide; Arc paralog 29328916, 39178227
Human disease	X-linked ASD (rare hemizygous LoF in males)	23352160; Ishino 2024

Evidence Base

PMID	Title (abbreviated)	Role in this report
26402067	*Cognitive Function Related to the Sirh11/Zcchc16 Gene Acquired from an LTR Retrotransposon in Eutherians*	**Cornerstone primary study.** Establishes Gag-derived CCHC identity, eutherian conservation pattern, the mouse KO cognitive phenotype, and the selective NA (not DA) microdialysis deficit.
28484174	*[Evolution of brain functions in mammals and LTR retrotransposon-derived genes]*	Review placing SIRH11/ZCCHC16 as an X-linked Gag-homologous CCHC zinc-finger gene whose deletion causes attention/impulsivity/working-memory defects "possibly via the locus coeruleus noradrenergic system," and notes lineage-specific structural variation.
37892118	*Retrovirus-Derived RTL/SIRH: Their Diverse Roles...*	Review defining the 11-member retrovirus-derived RTL/SIRH family context.
42511665	*Virus-Derived Domesticated Genes in Microglia and Resident Macrophages*	Supports RTL4's microglial expression and NA-responsive, stress-related brain role.
23352160	*Rare complete knockouts in humans... autism spectrum disorders*	Human exome framework: X-linked complete knockouts (including RTL4/ZCCHC16) contribute ~2% of male ASD.
38507667 / 37745568	*The Diverse Evolutionary Histories of Domesticated Metaviral Capsid Genes in Mammals*	Phylogenomics: SIRH/RTL are domesticated Metaviridae capsid genes; purifying selection + rapid evolution; diverse N-terminal domains.
29328916	*The Neuronal Gene Arc Encodes a Repurposed Retrotransposon Gag Protein...*	Paralog precedent: a domesticated Gag protein self-assembles into capsids, encapsulates RNA, and is released in EVs to transfer RNA — supports an analogous secretion/function route for RTL4.
39178227 / 38187623	*PI3P mediates Arc capsid secretion through the MVB pathway*	Mechanistic detail of the EV/MVB secretion route for Gag-derived capsids, informing RTL4's inferred unconventional secretion.

Convergence of evidence. The strongest line (mouse KO + microdialysis + Venus knock-in, all from the Ishino/Kaneko-Ishino group) is precise, low-throughput, and mechanistically consistent. Evolutionary genomics independently confirms the capsid-gene identity and selective constraint. Human genetics and Human Protein Atlas data provide convergent but lower-resolution support. The Arc paralog literature is used strictly as **structural/functional precedent**, not as direct evidence of RTL4 activity.

Limitations and Knowledge Gaps

1. **Human mechanistic data are essentially absent.** Nearly all functional evidence is from **mouse** *Sirh11/Zcchc16*. Human RTL4 is expressed at very low/near-threshold levels (HPA "Not detected" in bulk tissue and brain snRNA-seq), so no direct human biochemical or cell-biological characterization exists.
2. **Secretion route is inferred, not demonstrated for RTL4.** The EV/MVB/capsid pathway is established for Arc and partly for PEG10, not experimentally shown for RTL4. RTL4 may use a different unconventional route.
3. **No demonstrated molecular binding partner or substrate.** The CCHC knuckle predicts ssRNA binding, but RTL4's physiological interactors (RNA cargo, receptors, or extracellular partners) are unknown. The mechanism by which it "supports NA reuptake" is not resolved at the molecular level.
4. **Localization conflict.** HPA antibody-based imaging (nucleoplasm/plasma membrane/cytosol) contradicts the mouse secreted-protein data, reflecting likely antibody/overexpression artifacts at very low abundance. Protein-level localization in human cells is unresolved.
5. **ASD causality is suggestive, not definitive.** The human genetic link rests on rare LoF cases plus a statistical framework for X-linked knockouts; large-cohort replication specifically for RTL4/ZCCHC16 is limited.
6. **Capsid-assembly and RNA-transfer capacity are untested.** Whether RTL4 forms virus-like capsids or transfers RNA intercellularly (as Arc does) has not been assayed.

Proposed Follow-up Experiments / Actions

1. **Test unconventional secretion directly.** Express tagged human RTL4 in microglia/HEK293T and assay EV fractions (CD63+/MVB) for RTL4; test dependence on RalA/B, PI3P, and MVB machinery (mirroring Arc studies, PMID: 39178227).
2. **Assay capsid assembly and RNA encapsidation.** Purify recombinant RTL4, use negative-stain/cryo-EM to test for virus-like capsid formation, and perform CLIP/RNA-IP to identify any RNA cargo bound via the CCHC knuckle. Mutate the C-X2-C-X4-H-X4-C residues to test necessity.
3. **Define the noradrenergic mechanism.** In mouse, use conditional microglial Rtl4 deletion plus NA-reuptake (NET) and β-adrenergic-signaling assays to determine whether RTL4 acts on noradrenergic terminals, astrocytes, or NET function to promote NA reuptake.
4. **Identify binding partners.** Perform proximity labeling (BioID/TurboID) and co-IP/MS on secreted RTL4 to find extracellular or surface interactors in noradrenergic circuits.
5. **Strengthen human genetics.** Aggregate ZCCHC16/RTL4 LoF variants across ASD/ID exome databases (e.g., SFARI, DDD, gnomAD constraint) to formally test enrichment and X-linked constraint in males.
6. **Resolve the localization conflict.** Validate HPA antibodies with KO controls and use endogenous tagging in human stem-cell-derived microglia to establish bona fide human RTL4 localization.
7. **Peripheral catecholamine role.** Given adrenal-medulla enrichment, test whether RTL4 modulates catecholamine handling in chromaffin cells, which could offer a more tractable, higher-abundance human system than brain.

Conclusion

RTL4 (SIRH11/ZCCHC16/MART4; UniProt Q6ZR62) is a **eutherian-specific, X-linked domesticated retroviral Gag/capsid gene** whose 310-aa product retains an intact nucleocapsid-like CCHC zinc-knuckle in a basic, RNA-binding C-terminus but functions physiologically as a **secreted, microglia-derived, noradrenaline-responsive factor supporting the brain noradrenergic (locus coeruleus) system**. It acts extracellularly in noradrenaline-innervated regions to promote NA reuptake/noradrenergic network function

during neonatal critical periods; its loss impairs attention, impulse control, and working memory in mice, and rare hemizygous loss-of-function in human males is implicated in autism spectrum disorder. Because it lacks a signal peptide yet is secreted, it most likely reaches the extracellular space through an unconventional, capsid/extracellular-vesicle route analogous to its Gag-derived paralog Arc.
